

J-01-CX-geometric-depletion-kill-test v1: $W = |\omega|^2 \alpha$, $\alpha = \xi^T S \xi$,
no eigenframes

ZFL-CX

September 3, 2026

1 Mission

$$\boxed{T^* < \infty \Rightarrow C_J} \quad \boxed{C_E^{weak} \wedge C_{L^3} \wedge C_F^{conc} \wedge C_I^{freq} \wedge C_J \Rightarrow \int_0^{T^*} W_+ < \infty \text{ or } \perp} \quad W = \int |\omega|^2 \alpha dx, \quad \xi = \omega/|\omega|, \quad \alpha = \xi^T S \xi.$$

2 Attempt

$C_J^{Dir} : |\sin \varphi(\xi(x), \xi(y))| \leq |x - y|/\rho(t)$. If $\rho \geq \rho_0$ then $|S| \lesssim \rho^{-1}$, α_+ integrable. But under $T^* < \infty$, $\rho(t) \sim r_n \downarrow 0$ from $C_F^{conc} + C_I^{freq}$, so bound degenerates.

3 Kill

KILLED: only Constantin-Fefferman/Deng-Hou-Yu qualitative, needs $\rho_0 > 0$ not consequence of T^* , controls α pointwise not $|\omega|^2 \alpha$, no temporal integrability, compatible with $\int W_+ = +\infty$. Proven necessary: $C_J^{neg} : \limsup \|\nabla \xi\|_{L^\infty(\{|\omega| > M\})} = +\infty$ loss of direction regularity.